

The geography of intellectual capital and the potential onshoring of R&D: Evidence from TCJA Section 174

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Section 174 and the Tax Treatment of R&D

- TCJA (2017) was primarily deficit financed
- A pay-for: §174 in TCJA replaces expensing of R&D with amortization
- §174 set to go into effect for tax years beginning after **Dec. 31, 2021**
 - ▶ Business community expected repeal during each Congress 2018–2021 (Cowx et al. 2025)
- Extra §174 goal: onshoring innovative activity by making domestic R&D cheaper than foreign R&D
 - ▶ **Domestic R&D:** 5-year straight-line amortization $\implies$ $PV(\text{taxshield}) \downarrow \approx 14\%$
 - ▶ **Foreign R&D:** 15-year straight-line amortization $\implies$ $PV(\text{taxshield}) \downarrow \approx 29\%$

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Exploiting cross-firm heterogeneity in the
magnitude of the tax shock, driven by
pre-TCJA domestic vs. foreign R&D mix

Motivation and Research Question

Three big-picture questions:

- ➊ **Policy impact:** Did the 5 vs. 15-year wedge achieve its onshoring goal?
- ➋ **Elasticity:** How elastic are innovation outputs to the user cost of R&D?
- ➌ **Geographic substitutability:** Can domestic and foreign R&D inputs substitute for one another?

Motivation and Research Question

Three big-picture questions:

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Research Question

How did the **quantity, location, and characteristics** of innovation outputs at U.S. firms respond to an exogenous, firm-specific shock to the **user cost of R&D**?

Empirical Design: Three Measurement Challenges

$$Y_{igt} = \alpha_i + \gamma_{jt} + \beta (T_i \times Post_t) + \varepsilon_{it}$$

Problem 1: Unobserved internal R&D allocation (domestic vs. foreign)

- Necessary to define treatment status

Empirical Design: Three Measurement Challenges

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Problem 1: Unobserved internal R&D allocation (domestic vs. foreign)

- Necessary to define treatment status
- *Solution:* Pre-TCJA inventor location data (PatentsView, 2010–2016)

$$share_i = \frac{\# \text{ international inventors}_{i, 2010-2016}}{\# \text{ total inventors}_{i, 2010-2016}}, \quad T_i = 1[\text{share}_i \geq \text{median}]$$

Empirical Design: Three Measurement Challenges

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Problem 2: R&D allocation also unobservable as an outcome

- To measure impact of shock on innovative activity in places $\tilde{g} \in \{US, \text{Foreign}\}$

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Problem 2: R&D allocation also unobservable as an outcome

- To measure impact of shock on innovative activity in places $\tilde{g} \in \{US, \text{Foreign}\}$
- *Solution:* Measure allocation of invention outputs (patents) by inventor location
- $Y_{i,g,t}$ = outcome Y at firm i , for inventions of type g , in year t
- $g \in \{\text{Foreign-only, US-only, Collaborative}\}$, where “Collaborative” means at least one domestic and one foreign inventor on the same patent
- Y : count of distinct inventors, patents, technology classifications (“breadth”), and backward cites (“knowledge base”); sum of patent value

Empirical Design: Three Measurement Challenges

$$Y_{igt} = \alpha_i + \gamma_{jt} + \beta (T_i \times Post_t) + \varepsilon_{it}$$

Problem 3: Need output data through 2024

- Grants have multi-year lag; $\approx 50\%$ of 2024 innovations (applications) are not granted yet

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Problem 3: Need output data through 2024

- Grants have multi-year lag; $\approx 50\%$ of 2024 innovations (applications) are not granted yet
- *Solution:* Use data on patent *applications*
 - ▶ $\Rightarrow$ All 2024 applications were published by July 2025
 - ▶ Hurdle: Some data on applications requires scraping Google Patents
- Restriction: Hard to assess changes in "patent quality"
 - ▶ Forward citation data not available
 - ▶ We will use stock market reactions for *granted patents*

Empirical Design: Three Measurement Challenges

$$Y_{igt} = \alpha_i + \gamma_{jt} + \beta (T_i \times Post_t) + \varepsilon_{it}$$

Bonus Problem 4: Y has zeros

- : $Y_{i,g,t}$: count of distinct inventors, patents, technology classifications (“breadth”), and backward cites (“knowledge base”); sum of patent value
- These will contain zeros, especially for the foreign and collaborative categories
- Following Fitzgerald et al. (2026): poisson PML
- Will also show log1p and IHS specifications due to ease of interpretation

Sample Construction and Summary Statistics

Sample construction

- Compustat fiscal years 2019–2024, balanced sample (6 years each)
- Exclude utilities and financials (SIC 4900–4999, 6000–6999)
- Link patent applications to PERMNO/GVKEY via assignee ID
- To define T_i : 1+ US and 1+ foreign inventor during 2010–2016
- Active innovators: 1+ patent app in pre-period (2019–2021)
- Define treatment based on pre-TCJA inventor location share (2010–2016)
- Final sample: 801 firms (496 control, 305 treated)

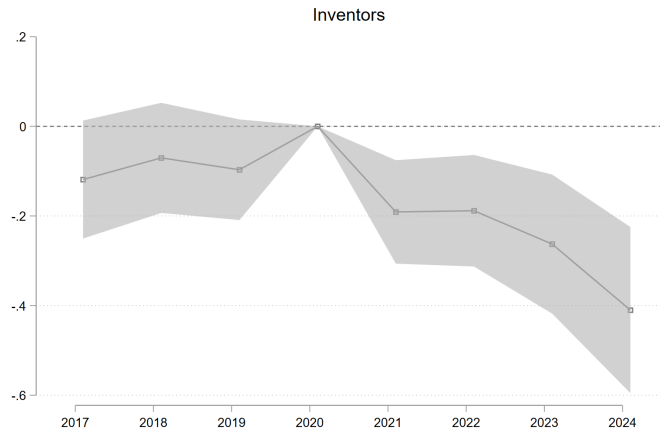
Sample Construction and Summary Statistics

	Control (T=0) mean	Treated (T=1) mean	Total mean
<i>Innovative Activities</i>			
NumInventors	47.05	137.65	81.56
PatApp	22.40	120.02	59.58
TechClasses	8.82	21.34	13.59
KnowBase	91.07	398.39	208.11
PatMktVal	1,596.54	1,527.80	1,570.36
<i>Financials</i>			
R&D	0.10	0.09	0.10
CAPX	0.03	0.03	0.03
Log(Assets)	7.58	8.47	7.92
Observations	2,975	1,830	4,805

The Innovative Activities of Foreign Inventors Fell

	(1) Inventors	(2) PatApp	(3) TechClasses	(4) KnowBase	(5) PatMktVal
Panel A: Poisson (PPML)					
$T_i \times \text{Post}_t$	-0.245*** (-2.62)	-0.187* (-1.69)	-0.194** (-2.52)	-0.376*** (-2.59)	-0.335* (-1.96)
Panel B: Log1					
$T_i \times \text{Post}_t$	-0.178*** (-3.33)	-0.146*** (-3.18)	-0.195*** (-4.76)	-0.418*** (-6.21)	-1.162*** (-11.67)
Panel C: IHS					
$T_i \times \text{Post}_t$	-0.208*** (-3.28)	-0.170*** (-3.12)	-0.234*** (-4.66)	-0.471*** (-6.07)	-1.286*** (-11.75)

The Innovative Activities of Foreign Inventors Fell

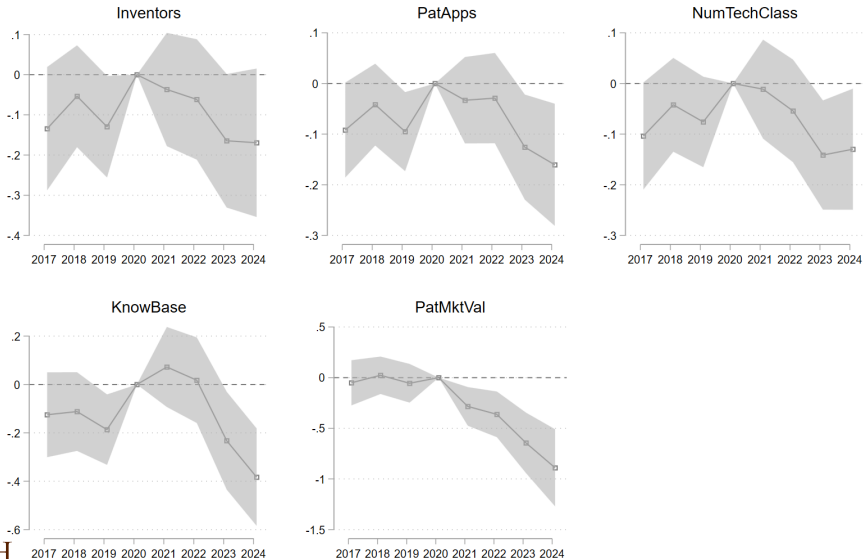


- Plotting Log1 model
- Including data to 2017 to better show pre-trends
- Other outcomes show similar patterns

Spillover: Collaborative Projects Declined

	(1) Inventors	(2) PatApp	(3) TechClasses	(4) KnowBase	(5) PatMktVal
Panel A: Poisson (PPML)					
$T_i \times \text{Post}_t$	-0.151* (-1.86)	-0.089 (-0.98)	-0.111* (-1.71)	-0.067 (-0.50)	0.296* (1.76)
Panel B: Log1					
$T_i \times \text{Post}_t$	-0.063 (-1.13)	-0.055 (-1.53)	-0.076** (-2.11)	-0.134** (-2.07)	-0.490*** (-4.69)
Panel C: IHS					
$T_i \times \text{Post}_t$	-0.073 (-1.08)	-0.066 (-1.48)	-0.092** (-2.03)	-0.148* (-1.94)	-0.546*** (-4.67)

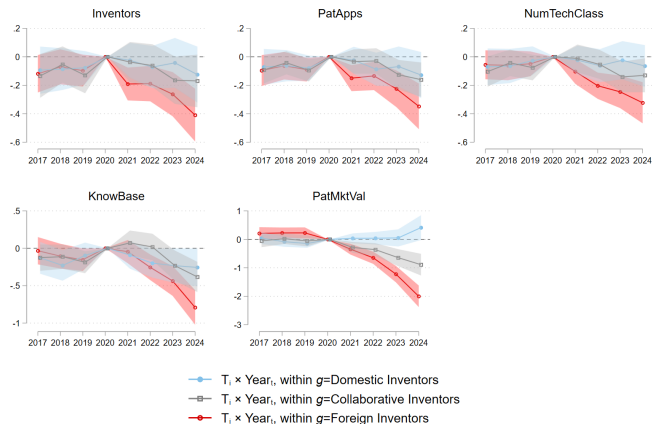
Spillover: Collaborative Projects Declined



Domestic Innovative Activity Did Not Offset the Decline

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$T_i \times \text{Post}_t$	-0.073 (-1.00)	-0.055 (-0.67)	-0.066 (-1.03)	-0.020 (-0.19)	0.050 (0.36)
Panel B: Log1					
$T_i \times \text{Post}_t$	-0.044 (-0.70)	-0.059 (-1.16)	-0.036 (-0.79)	-0.174** (-2.33)	0.166 (1.53)
Panel C: IHS					
$T_i \times \text{Post}_t$	-0.042 (-0.57)	-0.064 (-1.05)	-0.034 (-0.59)	-0.186** (-2.14)	0.208* (1.76)

Contrasting Responses by Inventor Location Within Firm

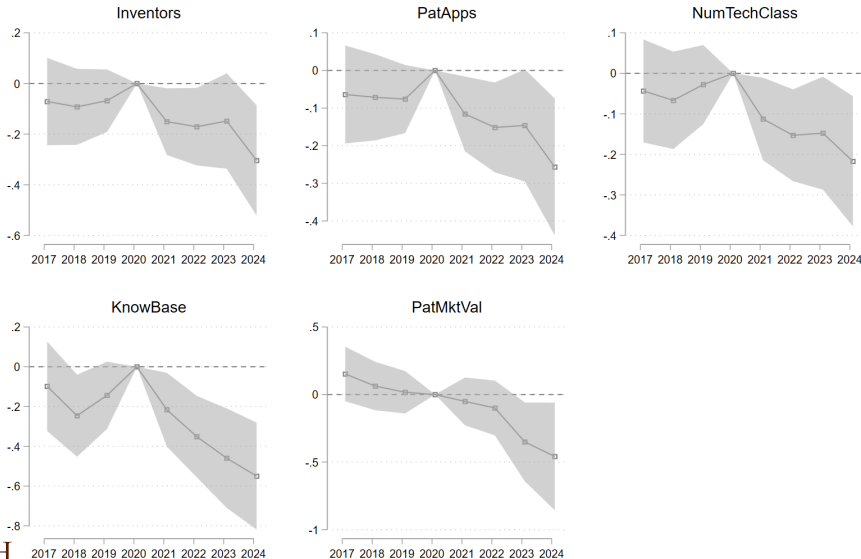


- All three groups in one stacked regression
- Rank order of TE: Foreign-only > Collaborative > US-only
- Visual DDD (exploits variation in across inventor groups within firm)
 - ▶ Parallel trends hold tightly
 - ▶ Effect on F inventors much stronger than effect on D inventors

Overall Innovation Outputs

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Overall Innovation Outputs



Innovation inputs (R&D) fall, especially for financially constrained firms

	(1) R&D	(2) CAPX	(3) R&D	(4) R&D	(5) R&D
$T_i \times \text{Post}_t$	-0.013** (-2.38)	0.001 (0.54)	-0.031** (-2.19)	0.001 (0.33)	-0.021** (-2.27)
$T_i \times \text{Post}_t \times \text{Neg Log(Asset)} (z)$			-0.017* (-1.84)		
$T_i \times \text{Post}_t \times \text{Neg DivPayer}$				-0.027** (-2.52)	
$T_i \times \text{Post}_t \times \text{EqtyCon} (z)$					-0.037** (-2.50)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year x Ind FE	Yes	Yes	Yes	Yes	Yes
Obs.	4,623	4,623	4,623	4,623	4,215
R^2	0.79	0.80	0.83	0.79	0.79

- 1 Decline in R&D higher for firms with larger R&D tax shock
- 2 Placebo test: no differential effect on CAPX (not subject to the policy)
- 3 R&D decline is larger for smaller, more financially constrained firms
- 4 → consistent with a **user-cost / cash-flow channel**
- 5 → a “first-stage” that can lead to reduced innovation output

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	(1) PatApp per Intl Inventor3	(2) PatApp per US Inventor3
Panel A: Poisson (PPML)		
$T_i \times \text{Post}_t$	0.308** (2.25)	0.006 (0.05)
Panel B: Log1		
$T_i \times \text{Post}_t$	0.034* (1.96)	-0.004 (-0.30)
Panel C: IHS		
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- *Productivity offset*: Intl inventors (col1) at T firms **increased** patents per inventor $\Rightarrow$ Consistent with endogenous cuts to marginal inventors
- Change to business interest deduction in 2022 (EBITDA $\rightarrow$ EBIT) did not confound results (robust to controlling for leverage and interest coverage)

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Contributions

- **Elasticity of R&D Location**

We provide a clean, within-firm estimate of how the geographic allocation of R&D responds to user cost, avoiding the endogeneity of cross-country panels or voluntary tax-credit take-up.

- **Domestic vs. Foreign R&D Substitutability**

We find that domestic and foreign R&D inputs are **not substitutes**. When the cost of foreign R&D rises, firms do not shift those projects to US inventors; they simply abandon marginal foreign and collaborative projects.

- **Financing of Innovation**

We provide causal evidence that tax-driven R&D cuts are amplified by financial constraints, linking tax design directly to corporate liquidity and the financing of innovation.

Conclusions: The Cost of Onshoring R&D

The Policy Goal: TCJA §174 aimed to onshore R&D by creating a massive user-cost wedge between domestic (5-year) and foreign (15-year) amortization.

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Thank you!

Questions?

<https://bowen.finance>